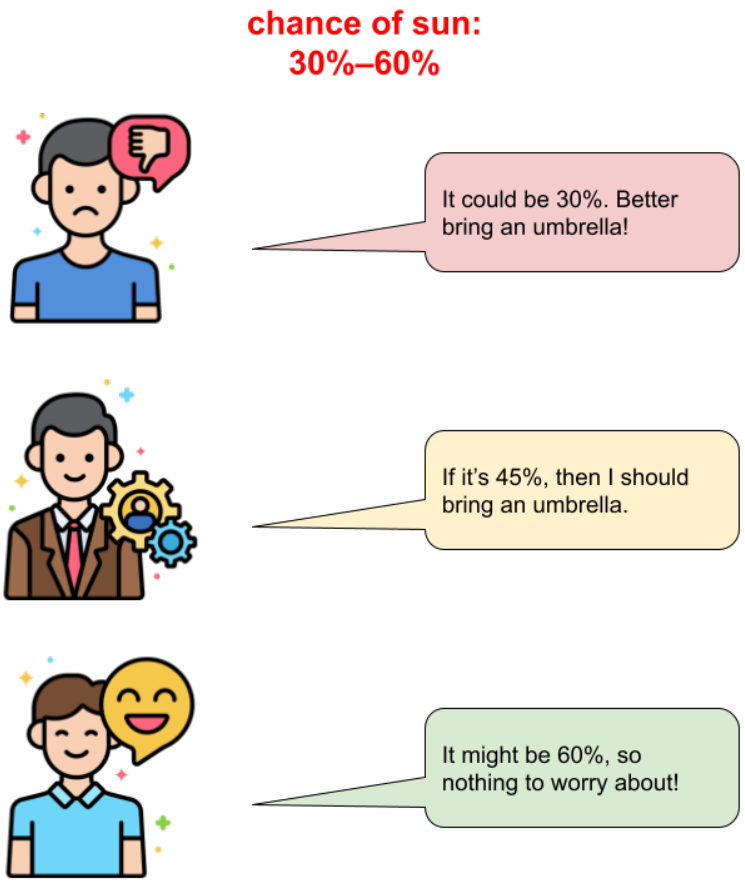


Figure 5.5 Different ways of thinking about the future and making decisions. The last person corresponds to the Upper Confidence Bound policy, which reasons about an unknown quantity in an optimistic manner.

> **NOTE** **Computing upper bounds with a Gaussian process**
>
> The way this policy is realized in BayesOpt is particularly simple, since with a Gaussian process as our predictive model of the objective function, we already have an upper bound of the reward rate of each action is. That is, we have an upper bound of the objective value at any given input location.

Specifically, we know that the objective value at a given location follows a normal distribution, and a commonly used measure to quantify uncertainty of a normal distribution is the 95% credible interval (CI), which contains 95% of the probability mass of the distribution. With a Gaussian process (GP), we visualize this 95% CI across the input space as the shaded region in Figure 5.6.

The upper bound of this 95% CI, which is the upper boundary of the shaded region highlighted in Figure 5.6, is exactly what the UCB policy uses as the acquisition scores for the data points in the search space, and the point that gives the highest score is the one at which we will evaluate the objective function next, which in this case is the location indicated by the dotted line around -1.3.

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